Mixture-Only 17-Class Comparison

Original real mixtures + PT2 real mixtures

Dataset. This comparison uses mixture data only: the 580-sample original real-mixture dataset plus the 324-sample PT2 real-mixture dataset, for 904 total mixture spectra. All scoring is still done in the full 17-class PT2-expanded inference space.

Important evaluation distinction. The tables in this document report final performance on real measured mixtures only. For the deep models, synthetic mixtures are used only to train the network and to run an internal synthetic train/validation/test model-selection loop. Those synthetic splits are not the numbers shown in Table 1 or Table 2. The reported benchmark rows are computed on the original real-mixture set and the PT2 real-mixture set.

Selection rule. "Replicate-Dictionary Pair NNLS" is chosen from the non-hand-tuned classical baselines on this 904-sample mixture-only set. "Similarity-Supervised Coefficient Regressor" is chosen from the non-hand-tuned pure-deep baselines on this same set. The localized classical fallback variants and the hybrid deep-plus-classical reranker are intentionally excluded.

Table 1: Full 17-class mixture-only comparison across original and PT2 real mixtures.

Model	Exact match	Micro-precision	Micro-recall	Micro-F1	Avg. returned chemicals
Siamese+MLP 2-head (default 0.5)	0.306	0.614	0.887	0.726	2.889
Siamese+MLP 2-head (calibrated)	0.301	0.572	0.905	0.701	3.166
Replicate-Dictionary Pair NNLS	0.940	0.970	0.970	0.970	2.000
Similarity-Supervised Coefficient Regressor	0.969	0.983	0.983	0.983	2.000

Table 2: Per-class precision, recall, and F1 on the combined mixture-only dataset. Classes with zero support across both mixture datasets are omitted.

Chemical	Support	Siamese 0.5			Siamese calibrated			Replicate-Dictionary Pair NNLS			Similarity-Supervised Coefficient Regressor		
		P	R	F1	P	R	F1	P	R	F1	P	R	F1
1-dodecanethiol	243	0.670	0.893	0.765	0.682	0.979	0.804	1.000	0.967	0.983	1.000	0.942	0.970
6-mercapto-1-hexanol	144	0.330	0.250	0.285	0.295	0.972	0.452	1.000	0.750	0.857	1.000	1.000	1.000
acetonitrile	72	0.667	1.000	0.800	0.667	1.000	0.800	1.000	1.000	1.000	1.000	1.000	1.000
benzene	301	1.000	1.000	1.000	1.000	0.880	0.936	1.000	1.000	1.000	1.000	1.000	1.000
benzenethiol	72	1.000	1.000	1.000	1.000	0.958	0.979	0.667	1.000	0.800	1.000	1.000	1.000
dichloromethane	108	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
diethylamine	36	0.279	0.944	0.430	0.279	0.944	0.430	0.800	1.000	0.889	0.706	1.000	0.828
etoh	121	0.606	0.992	0.752	0.723	0.818	0.767	1.000	1.000	1.000	1.000	1.000	1.000
meoh	243	1.000	0.872	0.932	1.000	0.901	0.948	1.000	0.959	0.979	1.000	0.934	0.966
n,n-dimethylformamide	72	0.889	1.000	0.941	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
n-hexane	108	0.727	0.667	0.696	0.727	0.667	0.696	1.000	1.000	1.000	1.000	1.000	1.000
pyridine	252	1.000	1.000	1.000	1.000	0.845	0.916	1.000	1.000	1.000	1.000	1.000	1.000
toluene	36	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Selected classical model family. Replicate-Dictionary Pair NNLS. Selected run: pair_nnls_replicate_dictionary. Candidate scores on the same mixture-only set: pair_nnls_replicate_dictionary: exact=0.940, micro-F1=0.970, pair_nnls_with_baseline_atoms: exact=0.928, micro-F1=0.964, exhaustive_pair_nnls: exact=0.913, micro-F1=0.956, cardinality_adaptive_nnls: exact=0.896, micro-F1=0.971, full_library_sparse_support_ exact=0.753, micro-F1=0.919.

Selected deep model family. Similarity-Supervised Coefficient Regressor. Selected run: deep_similarity_supervisio Candidate scores on the same mixture-only set: deep_similarity_supervision: exact=0.969, micro-F1=0.983, deep_binary_coefficient_regressor: exact=0.968, micro-F1=0.983, deep_cnn_encoder: exact=0.923, micro-F1=0.960, deep_replicate_decoder: exact=0.905, micro-F1=0.951.

How the default Siamese+MLP 2-head works. This is the original PT2-expanded retrained Siamese pipeline. It starts from synthetic binary mixtures generated from the expanded pure-spectrum library. Each synthetic sample mixes two reference compounds at sampled ratios, then passes the result through the same preprocessing used at test time. The model has two Siamese-style encoders: one branch for the preprocessed Raman spectrum and one branch for its FFT representation. Those two embeddings are concatenated and passed to a multilabel MLP presence head with one sigmoid output per compound in the 17-class library. Training therefore optimizes a class-presence detection problem, not an explicit unmixing problem. At inference, every class with output score above the fixed threshold 0.5 is returned as present, so the number of predicted chemicals can vary from sample to sample.

How the calibrated Siamese+MLP 2-head works. The calibrated Siamese model uses the exact same trained Raman encoder, FFT encoder, and multilabel presence head as the default row. The only change is the decision rule at inference time. Instead of using a flat 0.5 threshold for every class, it applies the saved class-specific thresholds that were tuned on the original real-mixture calibration experiment. That can improve recall for weak classes by lowering their trigger threshold, but it also increases the risk of overpredicting absent compounds because every class is still decided independently. In this mixture-only 17-class evaluation, that tradeoff pushes predicted support size up from about 2.89 to about 3.17 chemicals per sample and hurts exact-match performance.

How the replicate-dictionary pair NNLS model works. The selected classical winner is the replicate-dictionary pair NNLS model. It is not a trained neural network; it is a search-and-fit method built directly from the reference library. The pipeline starts from the original reference CSV and expands it with the PT2 pure spectra so inference is performed in the full 17-class library. Spectra are baseline-corrected before dictionary construction. For each compound, the method builds a small atom set rather than using a single prototype. That atom set contains the class mean spectrum plus up to nine representative replicate spectra selected to span within-class variation. A constant nuisance baseline atom is then appended so simple background offset can be absorbed without forcing one of the chemical atoms to explain it.

Replicate-dictionary pair NNLS inference. At test time the method assumes the mixture is binary, but it does not know which binary pair is present. It explicitly enumerates every candidate pair in the 17-class library, so the search covers 136 possible supports. For one candidate pair, the two compound-specific atom sets and the constant baseline atom are concatenated into a design matrix. Nonnegative least squares is then solved against the observed spectrum, so all coefficients are constrained to be nonnegative. This produces a reconstruction, a residual norm, and aggregate coefficient mass for the left compound, right compound, and baseline atom. The model repeats this for every candidate pair and returns the pair with the lowest residual. In other words, the inductive bias is enforced by the solver itself: binary support, nonnegative contributions, and a direct residual fit to real reference spectra.

What is and is not trained in the classical model. There is no learned parameter update from labeled mixtures. The only model-selection step is choosing a dictionary construction setting, such as the number of extra representative replicates per compound. In the selected run, baseline-corrected spectra and up to nine extra representative replicates per compound gave the strongest clean result. Once that dictionary is fixed, the method is entirely deterministic and prediction is driven by exhaustive pair search plus NNLS fitting.

How the similarity-supervised coefficient regressor works. The selected deep winner is the similarity-supervised coefficient regressor. It also uses the PT2-expanded 17-class reference library, but instead of searching over candidate pairs explicitly, it learns a direct mapping from one preprocessed spectrum to a 17-dimensional coefficient-share vector. The model is a multilayer perceptron operating directly on the spectrum after the same preprocessing used by the classical pipeline. Its architecture is three fully connected stages: input to 512 hidden units with layer normalization, GELU activation, and dropout; then 512 to 256 with layer normalization, GELU, and dropout; then 256 to 17 output logits. Those logits are passed through a softplus transform to force them positive and then normalized to sum to one, so the network predicts nonnegative share weights over all 17 compounds.

How the deep model is trained. The model is trained on synthetic binary mixtures generated from the expanded reference library, not on the real measured mixtures directly. For every pair of compounds in the 17-class library, synthetic mixtures are created at ratios from 0.1 to 0.9 in steps of 0.1, with ten random draws per ratio. Each draw uses random reference replicates from the two compounds, multiplies the mixed spectrum by a random global scale factor in the range 0.85 to 1.15, and adds Gaussian noise scaled to the spectrum's own standard deviation. The resulting synthetic set is split into train, validation, and test partitions stratified by pair identity, so the network is validated on held-out synthetic examples from every pair family before it is finally evaluated on the real measured mixtures.

Deep training objectives. Training does not use a single loss. First, there is a coefficient regression loss that tries to match the predicted shares to the true synthetic mixture ratios. Second, there is a support loss using binary cross-entropy on the 17 outputs so active compounds are separated from inactive ones. Third, there is a reconstruction loss: the predicted share vector is multiplied by a fixed decoder built from the normalized class-mean dictionary, and that reconstructed spectrum is pushed toward the input spectrum. The similarity-supervised variant then adds two extra terms beyond the baseline regressor. A margin-ranking loss forces the weakest true compound to remain above the strongest false compound by a margin. A similarity-weighted false-positive penalty uses the cosine similarity structure of the reference dictionary to punish false positives more heavily when they are spectrally close to the true compounds. That is the main mechanism aimed at reducing confusions such as chemically similar thiol-family competitors.

Deep inference. At inference, the deep model processes the spectrum once and returns a 17-way share vector. It does not threshold each class independently. Instead, it ranks the predicted shares and returns the top two compounds, which keeps the deployment rule matched to the binary real-mixture datasets while avoiding pair-specific hand tuning. So the deep model is globally learned but still uses a binary top-2 decision rule at the final step.

Why the similarity-supervised regressor wins here. On this combined mixture-only set, the similarity-supervised coefficient regressor outperforms the replicate-dictionary pair NNLS baseline because it keeps the same binary top-2 deployment rule while learning a smoother global ranking over the whole 17-class library. The classical NNLS solver is still very strong, but its remaining errors are concentrated in a few chemically similar neighborhoods where residual-based pair selection can flip to the wrong pair. The similarity-supervised regressor learns a library-level ordering that reduces those near-tie flips without using the hand-tuned fallback rules introduced only in the stronger engineering-only classical variants.